

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (IT)

May 2018

IT744 Information & Network Security

IT706.01/IT706 Information Security, Security Policies & Assurance

Date: 01.05.2018, Tuesday

Time: 01:30 pm To 04:30 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Explain the encryption of Hill cipher using suitable example. [07]

Q - 2 (a) Write a short note on algebraic group. [07]

(b) Prove that $2^{32} - 2$ is divisible by 171. [07]

OR

Q - 2 (a) Find the quadratic residues under Z_{19}^* . Which is (are) the primitive root(s) in the set? [07]

(b) Compute $21^{-1} \bmod 103$ using linear combination of 21 and 103. [07]

Q - 3 (a) Find the value of x using Chinese Remainder Theorem. [07]

OR

Q - 3 (a) Find the following: [07]

$\Phi(3), \Phi(20), \Phi(27), \Phi(127), \Phi(300)$

(b) Compute $21^{21} \bmod 31$ using fast exponential method. [07]

SECTION – II

Q - 4 Consider the following Frequency count (%) of Letters in English Text in some cipher text. [07]

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
11	6	11	3	5	5	6	13	8	4	7	5	12	7	2	8	14	6	7	9	8	18	10	8	9	3

Calculate index of coincidence value based on above table.

Q - 5 (a) Explain the working of kasiski test in vigenere cipher cryptanalysis attack. [07]

(b) Write a note on public key crypto system. Differentiate symmetric key cryptosystem and asymmetric key cryptosystem. [07]

OR

- Q - 5 (a)** Explain working of RSA of cryptosystem using suitable example. [07]
- (b)** Let $n = 299$ and $B = 5$ find one of the factor for given n using pollard $p - 1$ algorithm. [07]
- Q – 6 (a)** Find the factors of 8051 using pollard rho algorithm. [07]
- (b)** Explain the Soloway Stressen algorithm using suitable example. [07]

OR

- Q – 6 (a)** Consider prime field $GF(19)$, public key $[q, \alpha, \beta] = [2, 67, 94]$ and private key = $[67]$. [07]
Show
- (b)** Determine the Discrete Logarithm of 37 to the base 3 in Z_{101}^* using Shanks' algorithm. [07]
